From Stationarity to Feature Geometry

A residual-weighted refinement of NFA in two-layer networks

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github.com/elteomc/feature-learning

Background: two views of feature learning

NFA (Radhakrishnan et al., 2024)

An empirical ansatz:

$$B^\top B \;\propto\; \mathrm{AGOP}(f)^\alpha,$$

$$\mathrm{AGOP}(f) = \frac{1}{n}\sum_i \nabla f(x_i)\nabla f(x_i)^\top.$$

Strong empirical fit. General nonlinear theory is open.

FACT (Beaglehole et al.)

First-principles move: derive a feature self-consistency identity from

$$\nabla_W \mathcal{L} = 0 \quad \text{at convergence.}$$

Explains *when* NFA holds and *when* it can fail.

This project: a two-layer microscope

Take the FACT philosophy literally in

$$f(x) = a^\top \phi(Bx), \quad \mathcal{L} = \frac{1}{2n}\sum r_i^2 + \frac{\lambda}{2}\|B\|_F^2.$$

Expand the B-stationarity equation. Find:

- the natural object is *residual-weighted* AGOP, not raw AGOP.
- raw AGOP needs two extra links (beta, pair) which we identify and diagnose.

Contribution: a finite-sample decomposition of *when* raw NFA appears, *which* link breaks when it fails, and a partial empirical map.

What are we actually asking?

The NFA hope

The learned input metric $H := B^\top B$ should reflect input-gradient geometry:

$$G = \frac{1}{n} \sum_i \nabla f(x_i) \nabla f(x_i)^\top .$$

When is this true, and what breaks it?

The answer in this project

The stable late-training law is *not* $H^2 \approx G$. It is

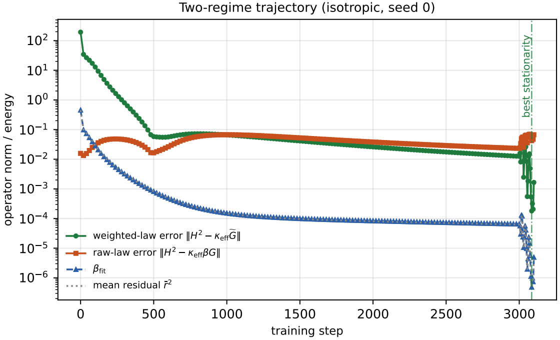
$$\boxed{H^2 \approx \kappa_{\text{eff}} \widetilde{G}},$$

$$\widetilde{G} = \frac{1}{n} \sum_i r_i^2 \, \nabla f_i \, \nabla f_i^\top .$$

Raw AGOP appears only when an additional *beta link* converts $\widetilde{G}$ into G, and that link becomes ill-conditioned near interpolation.

One-sentence thesis: *Raw NFA is regime-dependent. The primitive object is residual-weighted.*

Start with the phenomenon



Two-regime trajectory, isotropic seed. The step-3000 cliff is the SGD→L-BFGS hand-off (see right).

What you see

- Raw AGOP error (red) bottoms out in an *intermediate* window.
- Weighted-law error (green) keeps decreasing into stationarity.
- Beta and mean residual energy collapse together near interpolation.

The cliff at step 3000. There we hand off from SGD to a stronger optimizer (L-BFGS), which breaks the SGD plateau and drives the run *close to interpolation*. It is a real regime change, not a plotting glitch: as $r_i^2 \to 0$, the bridge scalar β_{fit} (a weighted average of the r_i^2) collapses with them, so de-weighting $\widetilde{G} \approx \beta G$ goes ill-conditioned while $H^2 \approx \kappa_{\text{eff}}\widetilde{G}$ stays accurate. **Implication.** A single end-of-training correlation

hides the trajectory. The weighted law is **stable late**, the raw law **conditionally visible**.

The story map: two stages, not one chain

Stage 1: primary spine (proved deterministically)

Stationarity ($\nabla_B \mathcal{L} = 0$) → High-gain pair closure → **Weighted law:** $H^2 \approx \kappa_{\text{eff}} \widetilde{G}$

Stage 2: optional bridge (regime-dependent)

Weighted law $H^2 \approx \kappa_{\text{eff}} \widetilde{G}$ → Beta link: $\widetilde{M} \approx \beta M$ → **Raw NFA-like law**

Pair closure fails $\Rightarrow$ *weighted* law fails.

Beta link fails $\Rightarrow$ *only raw* law fails, weighted law is unaffected.

Setup and the key stationarity identity

$$f(x) = a^\top \phi(Bx), \quad r_i = f(x_i) - y_i, \quad q_i = a \odot \phi'(Bx_i).$$
$$X = [x_1, \dots, x_n], \quad Q = [q_1, \dots, q_n], \quad R = \operatorname{diag}(r_1, \dots, r_n).$$

Loss and gradient

$$\mathcal{L} = \frac{1}{2n} \sum_i r_i^2 + \frac{\lambda}{2} \|B\|_F^2,$$

$$\nabla_B \mathcal{L} = \frac{1}{n} QRX^\top + \lambda B.$$

B-stationarity

$$\boxed{B = -\frac{1}{\lambda n} QRX^\top}$$

Residuals enter *automatically*. They are not a modeling choice, they are the loss derivative.

Defect form (used throughout)

Let $Z := \lambda n B + QRX^\top$. Then $Z = 0$ iff exactly stationary. We measure $\|Z\|$ at every checkpoint to quantify how close we are to the regime our theorem describes.

The key objects, in plain language

Two hidden-space matrices

$$
\begin{aligned}
T &= QR^2Q^\top && \text{(weighted hidden covariance)}\\
S &= QRX^\top XRQ^\top && \text{(same, with input pair geometry } X^\top X \text{ inserted)}
\end{aligned}
$$

- T is what the residual-weighted AGOP $\widetilde{G} = \frac{1}{n}B^\top TB$ is built from.
- S appears *automatically* from stationarity: $BB^\top = \frac{1}{\lambda^2 n^2}S$.
- **Pair closure** asks whether $X^\top X$ acts like a scalar on the sample combinations the network actually uses:

$$
d_{\text{eff}} := \frac{\langle S, T\rangle_F}{\|T\|_F^2}, \qquad \kappa_{\text{eff}} := \frac{d_{\text{eff}}}{\lambda^2 n}.
$$

Why d_{eff} deserves the name "effective dimension"

For isotropic X, $X^\top X \approx d\,I$ on relevant directions, giving $S \approx d\,T$. Adaptive training instead picks out a subset of directions, so the right scalar is the data and iterate dependent d_{eff}, not the ambient d.

Main result: a deterministic weighted square law

Four-line derivation (intuition)

$$\begin{aligned} \nabla_B \mathcal{L} = 0 \;&\Rightarrow\; B = -\tfrac{1}{\lambda n} QRX^\top \\ &\Rightarrow\; BB^\top = \tfrac{1}{\lambda^2 n^2}\, S \\ \widetilde{G} \;&=\; \tfrac{1}{n}\, B^\top T B \\ S \approx d_{\text{eff}} T \;&\Rightarrow\; H^2 \approx \kappa_{\text{eff}}\, \widetilde{G}. \end{aligned}$$

Theorem (deterministic, finite-sample)

Under exact B-stationarity,

$$H^2 - \kappa_{\text{eff}}\, \widetilde{G} \;=\; \frac{1}{\lambda^2 n^2}\, B^\top (S - d_{\text{eff}} T) B.$$

Hence $\|H^2 - \kappa_{\text{eff}} \widetilde{G}\|_{\text{op}} \leq \frac{\|T\|_{\text{op}}}{\lambda^2 n^2}\, \|B\|_{\text{op}}^2 \cdot \mathcal{A}_{\text{pair}}$, with $\mathcal{A}_{\text{pair}} = \|T^{\dagger/2}(S - d_{\text{eff}} T) T^{\dagger/2}\|_{\text{op}}$.

Important. What actually appears in the law is the *pushed* pair error $B^\top (S - d_{\text{eff}} T) B$, not the global $\mathcal{A}_{\text{pair}}$.

The two links, made precise

Link 1. Beta stability

$$\beta_{\text{fit}} = \tfrac{1}{n}\sum_i \ell_i\, r_i^2, \quad \tfrac{1}{n}\sum_i \ell_i = 1.$$

$$\frac{\beta_{\text{fit}}}{\bar{r}^2} - 1 \;=\; \mathrm{Corr}_n(\ell, r^2)\,\mathrm{CV}_n(\ell)\,\mathrm{CV}_n(r^2).$$

Holds when hidden-gradient leverage and residual energy are weakly correlated.

Fails when a few high-leverage samples carry systematically different residuals.

Collapses as $r_i^2 \to 0$ near interpolation.

Link 2. High-gain pair closure

Take SVD $QR = U\Sigma V^\top$. Define $G_{\text{stat}} := \Sigma^2 V^\top X^\top$ and $F_X := V^\top X^\top X V - d_{\text{eff}} I$. Then

$$B^\top (S - d_{\text{eff}} T) B \;=\; \tfrac{1}{\lambda^2 n^2}\, G_{\text{stat}}^\top F_X\, G_{\text{stat}}.$$

Holds when F_X is small on the high-gain directions of G_{stat}.

Note. Global $\|F_X\|_{\text{op}}$ can be large while the pushed error stays small. Global isotropy is a *conservative* diagnostic.

One-line summary

Beta failure $\Rightarrow$ *raw* law fails. Pair failure (high-gain) $\Rightarrow$ *weighted* law fails. These are different regimes.

Experiments: principled diagnostics across families

Positive families

- **Isotropic Gaussian**. Clean theory regime.
- **Anisotropic Gaussian**. Stresses the isotropy assumption.
- **Low-rank signal + isotropic noise**. Closer to realistic feature-learning data.

Adversarial families (constructed to break a specific link)

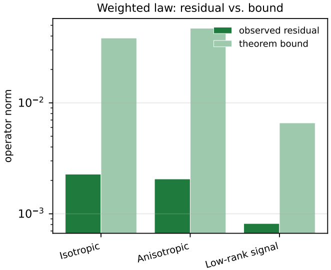
- Rare hard / rare easy high-leverage clusters.
- Two-region gating, mixture of subspaces.

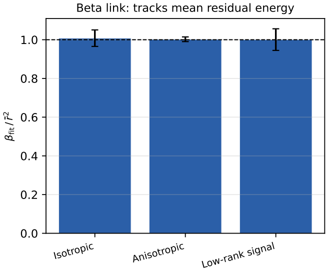
Five seeds per family, best-stationarity checkpoints.

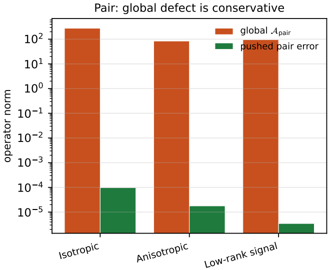
Diagnostics (theorem-aligned)

- Weighted-law residual $\|H^2 - \kappa_{\mathrm{eff}}\widetilde{G}\|_{\mathrm{op}}$. *Does the main law hold?*
- Theorem bound ratio. *Is the bound meaningful?*
- $\beta_{\mathrm{fit}}/\bar{r}^2$. *Does the beta link hold?*
- Pushed pair error $\|B^\top(S - d_{\mathrm{eff}}T)B\|_{\mathrm{op}}$ vs. global $\mathcal{A}_{\mathrm{pair}}$.
- Stationarity defect $\|Z\|_{\mathrm{op}}$.

Experiments: what we observe (positive families)







Lower is better.
Residual stays well under bound.

Near 1 is good.
$\beta_{\mathrm{fit}}/\bar{r}^2 \approx 1$.

Pushed error tiny.
Even when $\mathcal{A}_{\mathrm{pair}}$ is not.

Three messages

1. The **weighted law** is accurate at best-stationarity checkpoints.
2. The **beta link** holds in all three positive families ($\beta_{\mathrm{fit}}/\bar{r}^2 \approx 1$).
3. Global $\mathcal{A}_{\mathrm{pair}}$ can look scary while the **pushed** error remains small, exactly the conservative-diagnostic scenario the theorem predicts.

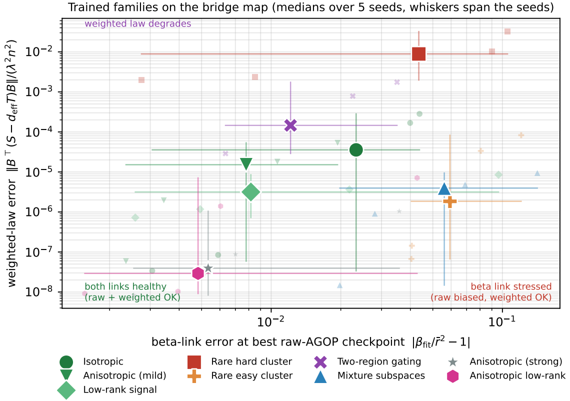
Predicted failure modes

Regime	Trigger	Consequence
Beta failure (hard rare cluster)	$\mathrm{Cov}_n(\ell, r^2) > 0$ large	raw law biased, weighted law fine
Beta failure (easy high-leverage)	$\mathrm{Cov}_n(\ell, r^2) < 0$ large	raw law biased the other way
High-gain pair failure	bad direction of F_X aligns with G_{stat}	**weighted law itself fails**
Conservative pair diagnostic	$\mathcal{A}_{\mathrm{pair}}$ large, $\lVert G_{\mathrm{stat}}^{\top} F_X G_{\mathrm{stat}} \rVert$ small	weighted law still holds
Late-regime collapse	$r_i^2 \to 0 \Rightarrow \beta \to 0$	raw conversion ill-conditioned

Closed-form sanity check (`run_failure_modes.py --toy`)

A deterministic $n = 40$ construction confirms the algebra: $\beta_{\mathrm{fit}}/\bar{r}^2$ moves from ≈ 1 (balanced) to $\gg 1$ (hard rare cluster) to ≈ 0 (easy rare cluster). The identity $\beta_{\mathrm{fit}} = \frac{1}{n}\sum \ell_i r_i^2$ checks to $< 10^{-10}$. *Next slide: a trained bridge map, and how robust the weighted law turns out to be.*

How robust is the weighted law? A trained bridge map



5 seeds per family. Marker = median, whiskers span the seeds. x: beta-link error $|\beta_{\mathrm{fit}}/\bar{r}^2 - 1|$ at the best raw-AGOP checkpoint. y: weighted-law residual $\|B^\top(S - d_{\mathrm{eff}}T)B\|/(\lambda^2 n^2)$. A guiding map, not a deliberate sweep.

What the map shows

- **The weighted law is hard to break.** Two pair-closure attacks built on purpose (strong anisotropy, anisotropic low-rank) sit bottom-left with the positives: near stationarity the network self-aligns, so the *pushed* pair error stays tiny even when global $\mathcal{A}_{\mathrm{pair}}$ is large.
- Mixture and rare-easy move *right*: the *beta* link is stressed (raw AGOP biased), the weighted law fine.
- Only the rare *hard* cluster moves *up* (two-region gating mildly): persistent amplified-outlier residuals (which also bend beta) are the one mechanism that degrades the weighted law itself.

Where things stand

Proved deterministically

- Exact B-stationarity identity.
- Residual-weighted square law $H^2 \approx \kappa_{\mathrm{eff}}\widetilde{G}$ with explicit error.
- Pushed-error factorization $\frac{1}{\lambda^2 n^2} G_{\mathrm{stat}}^\top F_X G_{\mathrm{stat}}$.
- Exact beta-bridge identity $\beta_{\mathrm{fit}} = \frac{1}{n}\sum \ell_i r_i^2$.
- **Bridge independence**: 2D toy instances exhibit each obstruction in isolation, so no universal NFA from stationarity alone.

Empirically supported

- Weighted law, beta tracking, pushed-pair phenomenon (5 seeds, 3 positive families).
- Trained map: the beta-failure modes appear as predicted, the weighted law proves robust to pair-geometry attacks, degrading only under persistent-residual structure (rare hard cluster).

Open

- *Dynamics route*: derive beta decorrelation from training (leverage-sensitive damping).
- *Adaptive concentration*: high-gain pair closure for learned subspaces.
- *Full phase diagram*: deliberate sweeps over leverage-residual correlation and high-gain anisotropy, with samplewise diagnostics.
- Extension beyond two layers (layerwise Jacobian factors).

Take-home

Raw NFA is useful but *regime-dependent*. The primitive object in two-layer stationarity is the *residual-weighted* law, and raw AGOP is a derived consequence governed by beta and high-gain pair closure.

Repo: `github.com/elteomc/feature-learning`